THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · EASY LEVEL MCQ TEST SET 2

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Easy level: direct, single-step questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. A vertical pole is 9 m high. The angle of elevation of the sun is 45° and the pole casts a shadow on level ground. The length of the shadow is: 1

(a) $9\sqrt{3}$ m **(b)** 9 m **(c)** 18 m **(d)** $\frac{9}{2}$ m

2. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 3 cm. The length of each tangent is 4 cm. The perimeter of the quadrilateral $PAOB$ is: 1

(a) 8 cm **(b)** 16 cm **(c)** 7 cm **(d)** 14 cm

3. The area of a circle is 154 cm^2. The radius of the circle is: 1

(a) 7 cm **(b)** 49 cm **(c)** 24.5 cm **(d)** 14 cm

4. A right circular cone has base radius 7 cm and height 24 cm. The curved surface area of the cone is: 1

(a) 704 cm² **(b)** 528 cm² **(c)** 550 cm² **(d)** 154 cm²

5. For the data with classes 20–30, 30–40, 40–50, 50–60, 60–70 with frequencies 3, 9, 4, 18, 16 respectively, The total number of observations $\sum f_i$ is: 1

(a) 47 **(b)** 50 **(c)** 49 **(d)** 51

6. A bag contains 3 red, 5 blue balls. One ball is drawn at random. The probability that it is red is: 1

(a) $\frac{3}{8}$ **(b)** $\frac{5}{8}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{1}{2}$

7. The prime factorisation of 60 is: 1

(a) 2×31 **(b)** 2×29 **(c)** $2^2 \times 3 \times 5$ **(d)** 2^6

8. A polynomial of degree 2 has at most how many zeroes? 1

(a) 1 **(b)** infinitely many **(c)** 2 **(d)** 3

9. The graph of $x = 2$ is a line parallel to the: 1

(a) line $y = x$ **(b)** y-axis **(c)** x-axis **(d)** none of the axes

10. The discriminant of $x^2 - 5x + 6 = 0$ is: 1

(a) -1 **(b)** 49 **(c)** 1 **(d)** 13

11. In an AP with $a = 5$ and $d = 3$, the 2nd term is: 1

(a) 8 **(b)** 11 **(c)** 2 **(d)** 15

12. The diagonal of a rectangle with sides 3 and 4 is: 1

(a) 1 **(b)** 5 **(c)** 25 **(d)** 7

13. The distance between the points $(6, 0)$ and $(0, 8)$ is: 1

(a) 14 **(b)** 100 **(c)** 2 **(d)** 10

14. $\sin 25^\circ = \cos$ of: 1

(a) 155° **(b)** 65° **(c)** 25° **(d)** 40°

15. From the top of a cliff 12 m high, the angle of depression of a boat on the sea is 45°. The horizontal distance of the boat from the foot of the cliff is: 1

(a) 6 m **(b)** 12 m **(c)** $12\sqrt{3}$ m **(d)** 24 m

16. PA and PB are tangents from an external point P to a circle with centre O, touching it at A and B. $\angle AOB = 110^\circ$. The $\angle APB$ is: 1

(a) 220° **(b)** 110° **(c)** 70° **(d)** 35°

17. A sector of a circle of radius 21 cm has central angle 60°. The area of the sector is: 1

(a) 22 cm² **(b)** 462 cm² **(c)** 1386 cm² **(d)** 231 cm²

18. A solid sphere has radius 7 cm. The surface area of the sphere is: 1

(a) 462 cm² **(b)** 616 cm² **(c)** 154 cm² **(d)** 308 cm²

19. For the data with classes 0–10, 10–20, 20–30, 30–40 with frequencies 3, 9, 5, 8 respectively, The cumulative frequency of the class 10–20 is: 1

(a) 9 **(b)** 25 **(c)** 12 **(d)** 3

20. 10 cards numbered 1 to 10 are placed in a box, mixed well and one is drawn at random. The probability that the number drawn is a multiple of 3 is: 1

(a) $\frac{7}{10}$ **(b)** $\frac{2}{5}$ **(c)** $\frac{1}{5}$ **(d)** $\frac{3}{10}$

21. The decimal expansion of $\frac{3}{8}$ is: 1

(a) non-terminating repeating **(b)** terminating
(c) non-terminating non-repeating **(d)** an integer

22. Which of the following is a zero of $x^2 - 5x + 6$? 1

(a) 5 **(b)** 2 **(c)** -2 **(d)** 1

23. The lines $x + 2y = 3$ and $2x + 4y = 6$ are: 1

(a) coincident **(b)** perpendicular
(c) intersecting at one point **(d)** parallel

24. The nature of the roots of $x^2 - 5x + 6 = 0$ is: 1

(a) two distinct real roots **(b)** two equal real roots
(c) no real roots **(d)** three real roots

25. The 1st term of an AP is 4 and the 3rd term is 10. The common difference is: 1

(a) $\frac{3}{2}$ **(b)** 3 **(c)** 6 **(d)** 7

26. Two triangles are similar if their corresponding angles are: 1

(a) supplementary **(b)** equal **(c)** in ratio 1:2 **(d)** complementary

27. The abscissa of the point $(5, 2)$ is: 1

(a) 2 **(b)** -5 **(c)** 7 **(d)** 5

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28. $\sec\theta$ is the reciprocal of: 1

(a) $\cot\theta$ **(b)** $\tan\theta$ **(c)** $\sin\theta$ **(d)** $\cos\theta$

29. A ladder 8 m long leans against a vertical wall and makes an angle of 60° with the level ground. The height reached on the wall is: 1

(a) 4 m **(b)** $8\sqrt{3}$ m **(c)** $2\sqrt{3}$ m **(d)** $4\sqrt{3}$ m

30. PA is a tangent at A to a circle with centre O, and $\angle OPA = 30^\circ$. The $\angle AOP$ is: 1

(a) 120° **(b)** 30° **(c)** 150° **(d)** 60°

31. A sector of a circle of radius 21 cm has central angle 60°. The length of its arc is: 1

(a) 44 cm **(b)** 64 cm **(c)** 231 cm **(d)** 22 cm

32. A solid sphere has radius 21 cm. The volume of the sphere is: 1

(a) 29106 cm³ **(b)** 5544 cm³ **(c)** 38808 cm³ **(d)** 19404 cm³

33. For the data with classes 10–20, 20–30, 30–40, 40–50 with frequencies 5, 8, 20, 17 respectively, The mean of the data is: 1

(a) 34.8 **(b)** 35.8 **(c)** 35 **(d)** 435

34. The probability of an event E is $\frac{1}{4}$. The probability of the event 'not E' is: 1

(a) $\frac{1}{8}$ **(b)** $\frac{5}{4}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{3}{4}$

35. The HCF of 20 and 30 is: 1

(a) 60 **(b)** 5 **(c)** 20 **(d)** 10

36. If $p(x) = 2x^2 + x - 5$, then $p(3)$ is: 1

(a) 16 **(b)** 11 **(c)** 21 **(d)** 10

37. The pair of equations $x + y = 5$ and $2x + 2y = 7$ has: 1

(a) no solution (parallel lines)
(b) infinitely many solutions (coincident lines)
(c) exactly one solution (intersecting lines)
(d) exactly two solutions

38. Which of the following is a quadratic equation? 1

(a) $-5x + 6 = 0$ **(b)** $x^3 - 5x + 6 = 0$ **(c)** $x^2 - 5x + 6 = 0$ **(d)** $x - 5 = 6$

39. The common difference of the AP $10, 7, 4, 1, \ldots$ is: 1

(a) -2 **(b)** -3 **(c)** 3 **(d)** -6

40. In a right triangle with legs 5 and 12, the hypotenuse is: 1

(a) 169 **(b)** 7 **(c)** 17 **(d)** 13

41. The distance of the point $(5, 12)$ from the origin is: 1

(a) 169 **(b)** 13 **(c)** 7 **(d)** 17

42. In a right triangle with perpendicular 12, base 5 and hypotenuse 13, $\sin\theta =$ 1

(a) $\frac{12}{13}$ **(b)** $\frac{13}{12}$ **(c)** $\frac{5}{13}$ **(d)** $\frac{12}{5}$

43. The line drawn from the eye of an observer to the point of the object being viewed is called the: 1

(a) horizontal line **(b)** tangent **(c)** line of sight **(d)** normal

44. A tangent to a circle meets the circle in exactly: 1

(a) one point **(b)** no point **(c)** three points **(d)** two points

45. A semicircular protractor-shaped plate has radius 7 cm. The perimeter of the plate is: 1

(a) 36 cm **(b)** 58 cm **(c)** 29 cm **(d)** 22 cm

46. A solid hemisphere has radius 7 cm. The total surface area of the hemisphere is: 1

(a) 616 cm² **(b)** 308 cm² **(c)** 462 cm² **(d)** 154 cm²

47. For the data with classes 20–30, 30–40, 40–50, 50–60, 60–70 with frequencies 3, 9, 4, 18, 16 respectively, The mode of the data is: 1

(a) 50 **(b)** 58.75 **(c)** 52 **(d)** 60

48. The probability of a sure event is: 1

(a) 2 **(b)** 0 **(c)** $\frac{1}{2}$ **(d)** 1

49. The LCM of 6 and 9 is: 1

(a) 54 **(b)** 3 **(c)** 15 **(d)** 18

50. The zero of the linear polynomial $3x + 9$ is: 1

(a) 9 **(b)** 3 **(c)** -3 **(d)** -9

END OF QUESTION PAPER **Total: 50 Marks**

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